

Advanced Machine Learning 2023

Final Exam

Chennai Mathematical Institute

04 December 2023, 09:30–12:30 (3 hours)

Marks: 100

Instructions:

- This question paper consists of two sides, with a total of 14 questions worth 100 marks. All questions must be answered; there are no optional ones.
- Before you begin writing your answers, carefully review all the questions.
- Make sure to write you name and roll number on the main booklet *and* on all the additional sheets that you use.
- If you need to use the restroom, ask for a toilet pass from the invigilators.
- Attendance will be taken during the exam. An invigilator will approach you with the attendance sheet. Please sign next to your name.
- **WARNING:** CMI's academic policy regarding cheating applies to this exam. Any form of copying or use of electronic devices is strictly prohibited. Violation will result in an automatic F grade, immediate exclusion from placement activities, and potential expulsion without exception.

1. Consider a neural network with 1 input, 1 output and k hidden layers, each with d neurons with ReLU activation. Calculate the total number parameters of this network. (5 marks)
2. Describe the activation functions Sigmoid, ReLU and Leaky-ReLU with leak α . Plot the curves of their gradients (approx values) in the range $[-10, 10]$. From these values what can you say about their effects on the gradient descent algorithm in a neural network. (5 marks)
3. Consider a fully-connected neural network for classifying images (say MNIST). What happens if we initialize all parameters of a neural network to zero at the start of training? What happens if we set them all to one instead? (5 marks)
4. Recall that we studied 2D convolutions in class. Can you describe a 1D convolution of kernel size 3 with 5-channel input. Write down the mathematical equation for this convolution. (5 marks)
5. Suppose you have a simple neural network, that given a number x as input, computes a number y as described below. Write down the equivalent residual network for this neural network. (5 marks)

$$x \longrightarrow z_1 = h_1(x) \longrightarrow z_2 = h_2(z_1) \longrightarrow y = h_3(z_2)$$

6. Consider the collection of triplets of intervals of the real-line $\{[a, b] \cup [c, d] \cup [e, f]\}$. What is the VC dimension of this collection? (5 marks)

7. Describe the regularizer term in the loss function for Variational Autoencoder. Explain how it affects the latent space, e.g. when decoding a random point or interpolating between two points corresponding to two different images in the training set. (5 marks)
 8. Describe how can compute the Action-value function from the state-value function of an MDP, and vice-versa. (5 marks)
 9. What is "Experience Replay" in Deep Reinforce Learning and why is it necessary? What happens if we do not incorporate Experience Replay in the training process? (5 marks)
 10. What is the key difference between Monte-Carlo methods and Temporal Difference Learning methods in Reinforcement Learning? (5 marks)
 11. Describe a small neural network with ReLU activations that computes (1) the average of four numbers and (2) the maximum of four numbers? Can you use this to implement 2x2 MaxPooling and 2x2 average pooling? (10 marks)
 12. Describe an LSTM unit in full details including the functions of it's various gates, inputs and outputs. (10 marks)
 13. Suppose you are training a GAN to generate images of cats and dogs. After training for sometime you take a look at the output of the Generator and observe that it is producing very good quality images of both cats and dogs. However, the discriminator still is able to successfully identify fake images at least 50% of the time. Can you give an explanation for this scenario. (10 marks)
 14. Suppose that you are developing a deep learning model for a client (e.g. an airline) that is given a short social media posts (like a tweet) as input and you must classify it is positive, negative or irrelevant to the client. The input will be short english sentence (at most 50 words) and the output must be one {positive, negative, irrelevant}. You will be given a training-set of 10000 labeled sentences. Please describe a complete deep learning model for this task, using 1D Convolutions. Also describe how you will train and test this network using the given data. (20 marks)
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Chennai Mathematical Institute

Regression and Classification

Final Exam: Aug-Nov, 2023

Total Marks: 80 points

Total Time : 3 hours

Attempt all problems.

Problem 1 (10 points)

- (1 point) In logistic regression, what distinguishes the probit link function from the logit link function?
- (2 points) Briefly explain the relationship between cross-entropy and the negative log-likelihood function.
- (2 points) Why will the performance of linear discriminant analysis and logistic regression be equivalent for the same feature space?
- (2 points) Define distributed multinomial logistic regression for a k -class classification problem.
- (3 points) For categorical time series data $\mathcal{D} = y_t : t = 1, 2, \dots, T$, where $y_t = 0$ with probability p_t and $y_t = 1$ with probability $(1 - p_t)$, develop an auto-regressive time series model of order p within a logistic regression framework.

Problem 2 (10 points)

The linear discriminant rule for class k is presented as:

$$\delta_k(x) = x^T \Sigma^{-1} \mu_k - \frac{1}{2} \mu_k^T \Sigma^{-1} \mu_k + \log \pi_k,$$

the best decision rule is

$$G(x) = \underset{k}{\operatorname{argmax}} \delta_k(x).$$

Consider the following summary statistics from two (independent) data sets:

$$\bar{X}_1 = \begin{pmatrix} 2 \\ 3 \end{pmatrix}, \quad \bar{X}_2 = \begin{pmatrix} 5 \\ 7 \end{pmatrix}, \quad S = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix} \quad n_1 = 20, \quad n_2 = 40$$

- (5 points) Compute the linear discriminant function. Use that linear discriminant function to classify the point $x = [1 \ 4]$ as either π_1 or π_2
- (5 points) Under which theoretical assumptions do you expect your method to be reliable?

$$\log(2) = 0.693 \text{ and } \log(3) = 1.099$$

Problem 3 (10 points)

- a) (3 points) Briefly explain when the Gaussian process prior regression model is useful
- b) (4 points) Modify the Gaussian process prior regression for a binary classification problem.
- c) (3 points) Why does the Gaussian process prior model fail for big data?

Problem 4 (10 points)

- a) (3 points) Briefly explain the relationship between basis expansion technique in functional regression and feature engineering in machine learning.
- b) (2 points) Briefly explain Kosambi-Karhunen-Loève theorem in the context of Gaussian process prior.
- c) (2 points) Explain the relationship between the distance between points and squared-exponential covariance function of Gaussian process model.
- d) (3 points) Consider the following statement.

Gauss-Markov Linear Models are special cases of Gaussian Process Regression Models.

Prove or Disprove the above statement.

Problem 5 (10 points)

- a) (3 points) What is tree structured regression?
- b) (3 points) What is appropriate objective function for tree structured regression? How to fit a tree-structured regression?
- c) (4 points) What is intrinsic feature selection? How tree structured regression achieves intrinsic feature selection?

Problem 6 (10 points)

- a) (2 points) How non-monotonic relationship can be captured in logistic regression?
- b) (2 points) Briefly explain the relationship between logistic regression and neural network.
- c) (2 points) The solution for Ridge regression is presented as

$$\hat{\beta} = (X^T X + \lambda I)^{-1} X^T y$$

- where $\lambda > 0$ is unknown. How can you estimate λ ?
- d) (4 points) Why Fourier Basis expansion does not suffer from multicollinearity problem?

Problem 7 (10 points)

Equations of the lines of regression of Y on X is

$$Y - \bar{Y} = r \frac{s_y}{s_x} (X - \bar{X}),$$

and equations of the lines of regression for X on Y are:

$$X - \bar{X} = r \frac{s_x}{s_y} (Y - \bar{Y}).$$

Note that

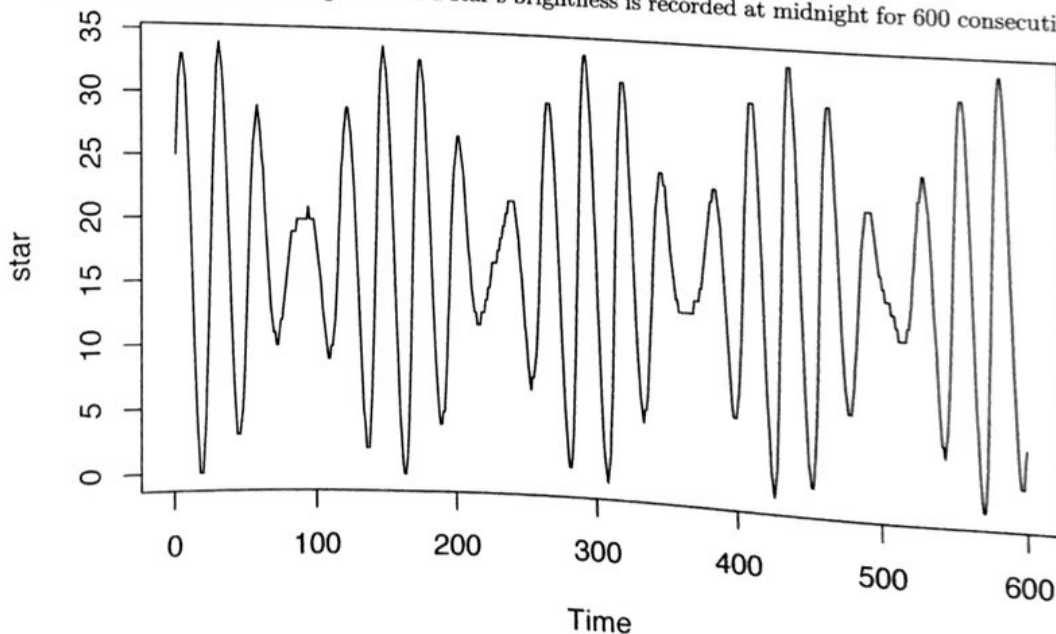
- $\bar{X}, \bar{Y}$ are the sample mean of X and Y ,
- s_x is the sample standard deviation of X
- s_y is the sample standard deviation of Y
- r is the sample correlation coefficients between X and Y .

- (4 points) If θ is the acute angle between the two regression lines then obtain θ in terms of r, s_x, s_y .
- (3 points) If $r = 0$ then find θ ?
- (3 points) If $r = 1$ then find θ ?

Problem 8 (10 points)

A variable star is a star whose brightness, as observed from Earth, fluctuates over time. These variations can occur for various reasons, leading to the classification of variable stars into different types based on the nature and cause of their brightness changes.

In the following graph, the magnitude of a star's brightness is recorded at midnight for 600 consecutive days.



- (4 points) Propose an appropriate regression model for predicting the brightness of a star.
- (4 points) Propose an alternative regression model for consideration.
- (2 points) What would be your strategy for identifying the best and second-best models?